

SEMESTER END EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E. - Information Science and Engineering	Semester	: IV
Course Name	: Microcontrollers	Max. Marks	: 100
Course Code	: IS44	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- List and explain the four major design rules in implementation of RISC processors. CO1 (08)
 - With a neat diagram, explain the internal components of a typical Microcontroller unit. CO1 (06)
 - Explain how the ARM instruction set differs from the pure RISC instruction set? CO1 (06)
- With a block diagram explain the characteristics of an embedded system. CO1 (08)
 - List the two modes of data transfer between MCU and peripherals? Explain the general functioning of interrupts. CO1 (06)
 - Describe the working of power on reset signal with a circuit diagram and waveform. CO1 (06)

UNIT - II

- With the timing diagram, explain memory read and write cycles of an SRAM chip. CO2 (08)
 - How is power management taken care at design stage? Explain Thermal Design Power (TDP). CO2 (07)
 - What is Direct Memory Access? Explain the typical setting of a DMA with neat diagram. CO2 (05)
- Differentiate between Inter Integrated Circuit (I2C) bus and Serial Peripheral Interface (SPI) Bus. CO2 (08)
 - Explain the current program state of an ARM7TDMI if the CPSR had the value 0xF00000D3. CO2 (07)
 - What is an Instruction Set Architecture (ISA)? Illustrate the structure of a typical ISA. CO2 (05)

UNIT - III

- How does Arm processor handle Interrupts? List vectors of predefined interrupts. CO3 (08)
 - List and explain seven optional features of ARM processor. CO3 (07)
 - Write an ARM assembly code to sort n bytes using bubble sort technique. CO3 (05)

6. a) With a block diagram, explain the various components of an ARM processor. CO3 (05)
- b) AREA Prog1, CODE, READONLY CO3 (07)
ENTRY
MOV R0, #0x11
LSL R1,R0, #1
LSL R2, R1, #1
stop B stop
END
i. What are the contents of R1 and R2 after the execution of the code?
ii. Replace the last LSL instruction by ADD R2, R1, R1, LSL #2
What is the value of R2 after the new execution?
- c) Find faults in the following instructions, if any and suggest a remedy: CO3 (08)
i. MOV R0, #300
ii. ROL R0, #4
iii. LDMH R0, {R1-R5}
iv. MUL R0, R0, R15.

UNIT- IV

7. a) Write an ALP to compute sum of squares of 5 numbers starting from 1. CO4 (10)
Write and use procedure SQU. Store sum in register.
- b) Explain the internal BUS structure of the LPC 214X families with appropriate diagram. CO4 (10)
8. a) Write an ALP to count the number of times a substring is repeated in the string. CO4 (08)
- b) Write a C program to display the string 'I LOVE ISE' in the serial window of UART1. CO4 (08)
- c) Define Load and Store instructions in ARM programming with example. CO4 (04)

UNIT - V

9. a) Discuss with appropriate diagram for a memory map model for the cortex-M0 processor. CO5 (10)
- b) Discuss the advantages of choosing Cortex M0 by the user for design of a processor. CO5 (10)
10. a) Briefly discuss on the use of NVIC in the cortex processor, also list its features. CO5 (10)
- b) Discuss on three-stage pipeline for Cortex-M0 & Two-stage pipeline for cortex-M0+ with appropriate diagram for each. CO5 (10)
